RITAM MUKHERJEE

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EDUCATION

University of Waterloo — Expected May 2030
Bachelor of Applied Science in Mechatronics Engineering — *Waterloo, Ontario*
- **Relevant Coursework:** Data Structures and Algorithms (C++), Digital Logic, Linear Algebra

EXPERIENCE

Independent Research Project – Dr. Eugene Li — Aug 2026 – Present
Undergraduate Student Researcher — *Waterloo, Ontario*
- Building a multi-camera pose estimation pipeline fusing synchronized camera streams via YOLOv8 and transformers; modeled BSIP anthropometrics and dynamic kinematics (CoM, ground reaction forces, 3D joint trajectories).

WATonomous — Sep 2026 – Present
Core Member - Humanoid Perception — *Waterloo, Ontario*
- Building ROS 2 (C++/Python) nodes synchronizing three camera streams with 100 Hz joint telemetry via ApproximateTime, recording teleoperated demonstrations to rosbag2/MCAP at 30 Hz for imitation-learning policy training.
- Handling multi-minute long-horizon manipulation episodes with clock-drift correction across sensor sources and segmentation into labeled subtask phases.

SuraNeoAge Systems Transformations (Early Stage Startup) — May 2026 – Aug 2026
Software Engineering Intern — *Remote*
- Fine-tuned Phi-4 Mini (3.8B params) via QLoRA for financial entity and topic extraction, using GPT-4o-generated ground-truth labels across 2,700+ scraped financial news articles enriched with National Stock Exchange sector data; outperformed the base model and GPT-4o mini in a 100-article blind ABC comparison judged by the team's business analyst.
- Built the ingestion-to-training data pipeline (PostgreSQL, FastAPI, Docker) and fine-tuned FinBERT (110M params) for company NER, reaching 0.76 token F1 / 0.88 recall.
- Built an on-demand LLM relevance-scoring endpoint replacing manual editorial curation; persisted latest verdict per article in Redis with automatic cache invalidation on reclassification, decoupling read latency from LLM inference time.

University of Waterloo – NanoTech lab and RoboHub — Sep 2025 – May 2026
Undergraduate Student Researcher — *Waterloo, Ontario*
- Worked under Dr. Shirley Tang on early-stage development of a 7-axis 3D printer using a Gen3 robot arm; implemented ROS 2 Cartesian trajectory-tracking and homing scripts on the physical arm, while researching non-planar slicing and g-code conversion.

University of Waterloo – WATiMAKE lab — Aug 2024 – Jun 2025
Volunteer Student Researcher — *Waterloo, Ontario*
- Assisted Dr. Eugene Li on an industrial CNC tool remaining-useful-life project; built a sensor-to-model data pipeline, prototyped a PyTorch Transformer, and trained an NCA+KNN classifier reaching 79.6% test accuracy on 1.5M+ accelerometer samples.

PROJECTS

Axiom | *GitHub* | *Node.js, TypeScript, Supabase, Vercel*
- Built an AI platform to standardize Canadian school grades via test-difficulty analysis, a 3D interactive map, and a natural-language agent.
- Developed full-stack with Next.js, Gemini AI, Mapbox, and Supabase with crowd-sourced adjustment data persistence.

Chess Engine - ResNet CNN + Search | *GitHub* | *Python, CNNs, Numpy, HuggingFace*
- Built a chess engine using a ResNet style CNN, policy/value head, and a depth-3 search to generate moves within 250ms.
- Used Modal cloud computing to train the model, uploading model weights and storing the model on HuggingFace.

Personal Website | *ritammukherjee.ca* | *Next.js, React, TypeScript, Supabase, Vercel*
- Developed a full-stack portfolio featuring live integrations with Spotify and Discord APIs to display real-time user activity.
- Implemented a custom messaging system using Supabase for the backend and integrated GitHub data to showcase dynamic project updates and repository statistics.

Motion Activated Door Monitoring System | *GitHub* | *Raspberry Pi, OpenCV, Docker, Ngrok, FastAPI*
- Developed a real-time motion detection system using OpenCV on a Raspberry Pi to monitor a webcam feed and detect door activity.
- On detection, activated a display, streamed live video, played alert audio, enabled two-way audio, and uploaded images to FastAPI via Ngrok.

TECHNICAL SKILLS

Languages: Python, C++, C#, TypeScript, JavaScript, SQL
Machine Learning: PyTorch, HuggingFace, scikit-learn, Numpy, OpenCV, TensorFlow, Pandas
ML Methods: Deep Learning, Computer Vision, NLP, LLM Fine-Tuning (QLoRA/LoRA), Transformers, Reinforcement Learning
Robotics: ROS 2, Foxglove Studio, Kinematics (FK/IK), Trajectory Planning, State Estimation (EKF/UKF), Sensor Fusion, Teleoperation, SolidWorks
Web/Infra: FastAPI, Next.js, React, PostgreSQL, Redis, Supabase, Docker, Vercel, Git, Bash, Linux, REST APIs, CI/CD